

Technology Stack

Date	30 September 2026
Team ID	SWTID-2026-6111
Project Name	FIT BUDDY AI
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS, Jinja2	Provides the user interface and dynamic templates for user input, workout results, feedback, and admin views.

2	Backend / Server-Side	Python + FastAPI (Uvicorn)	Handles routing, form input, validation, AI calls, and server-side application logic.
3	Database / Data Storage	SQLite + SQLAlchemy	Stores user details and workout plans; SQLAlchemy provides ORM-based persistence.
4	Cloud / Hosting / Deployment	Local deployment (Uvicorn)	Runs the FastAPI application locally for development and testing.
5	Version Control & CI/CD	Git	Used for version control as part of the development environment.

	Third-Party APIs / Other Tools	Google Gemini API (Gemini 1.5 Pro + Gemini Flash)	Generates personalized 7-day workout plans, updates plans from user feedback, and provides nutrition or recovery tips.
6			